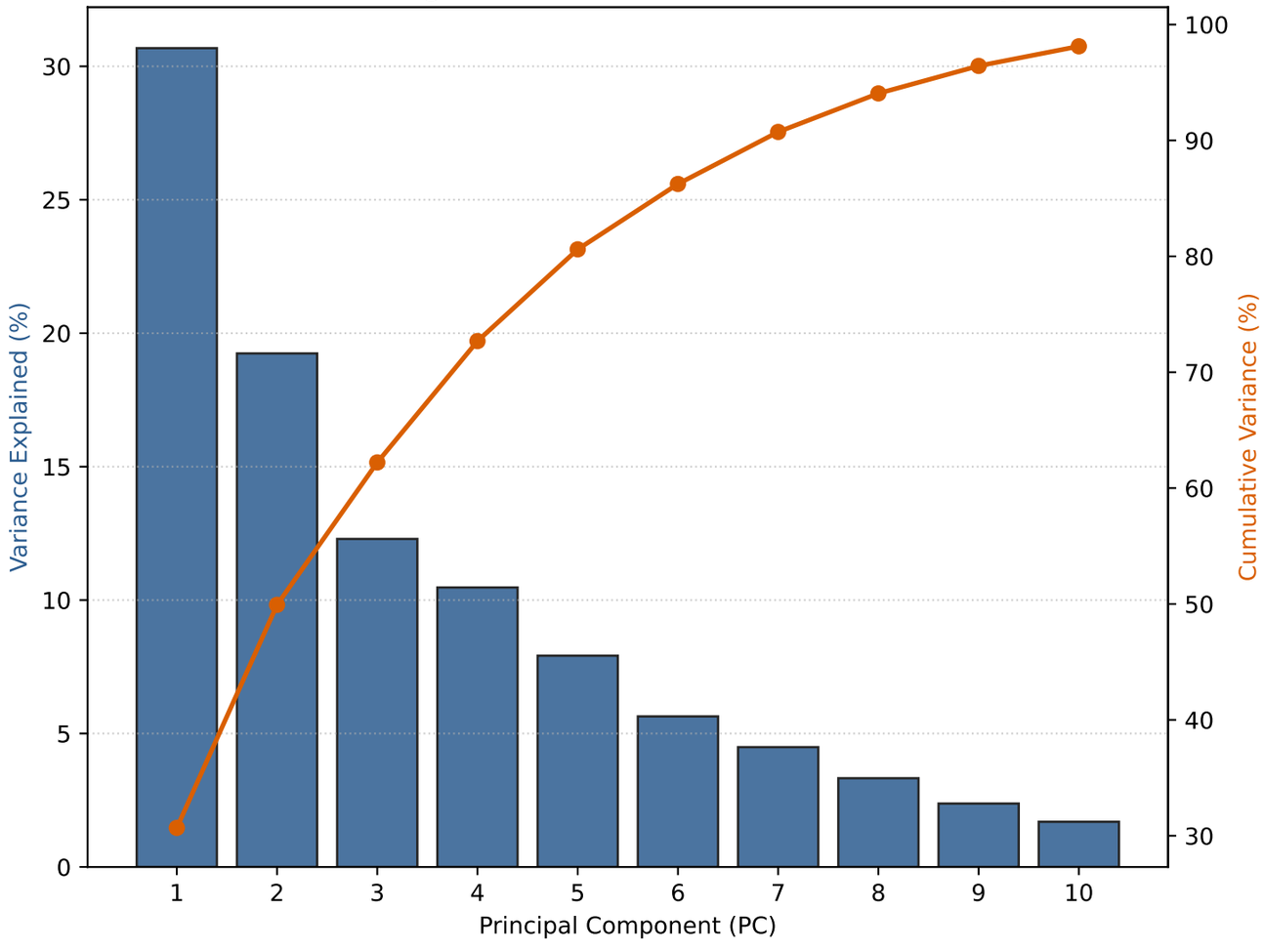
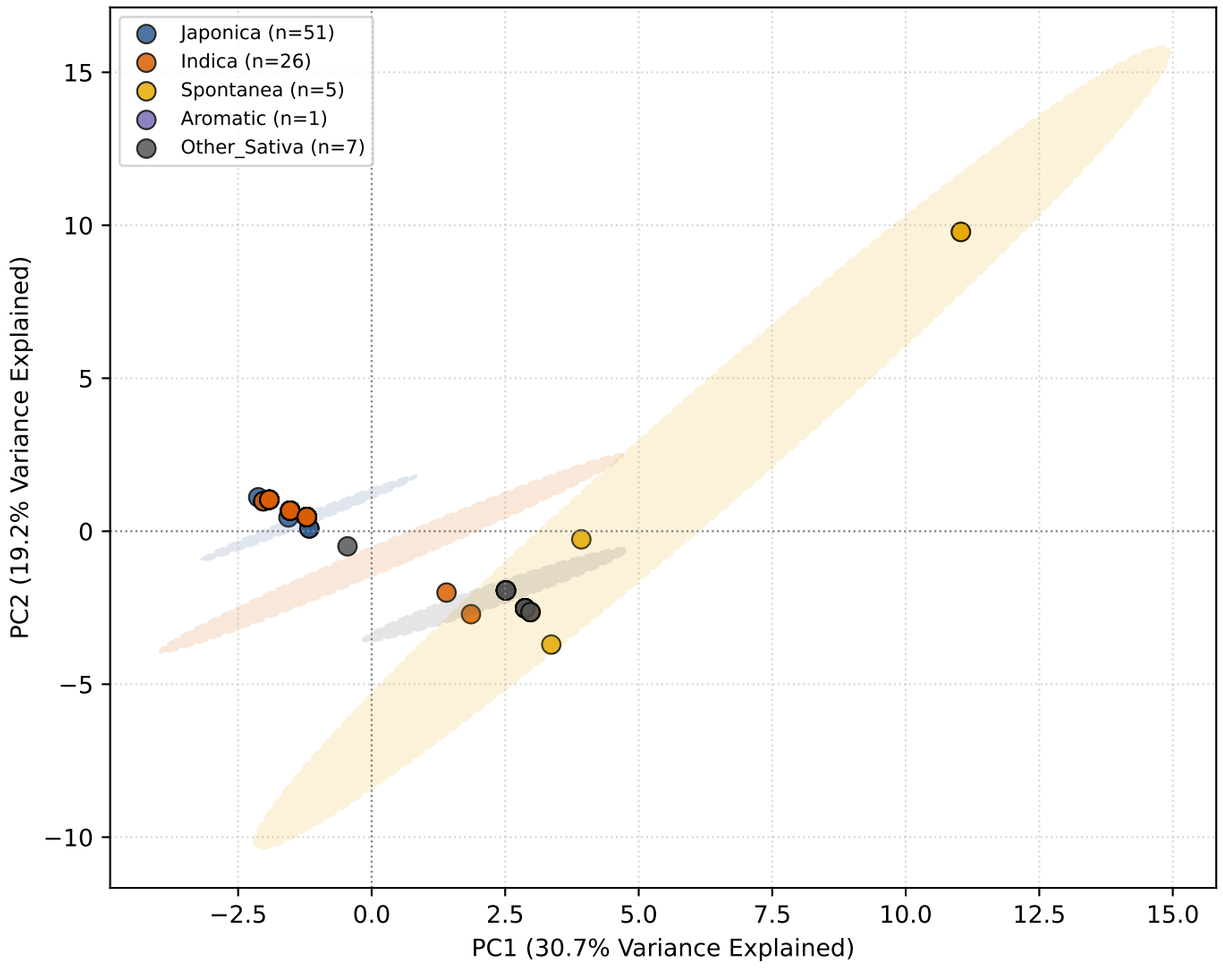
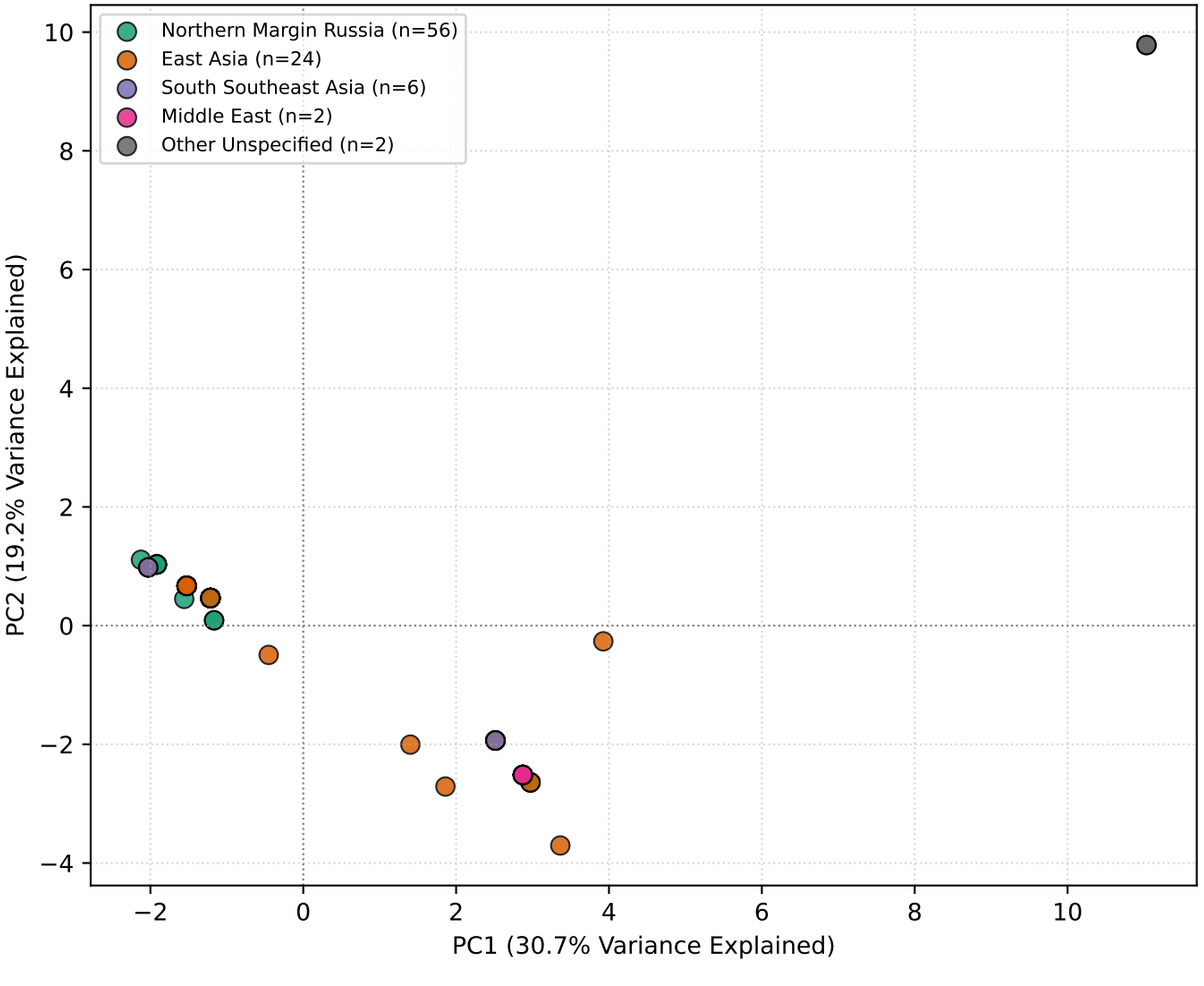
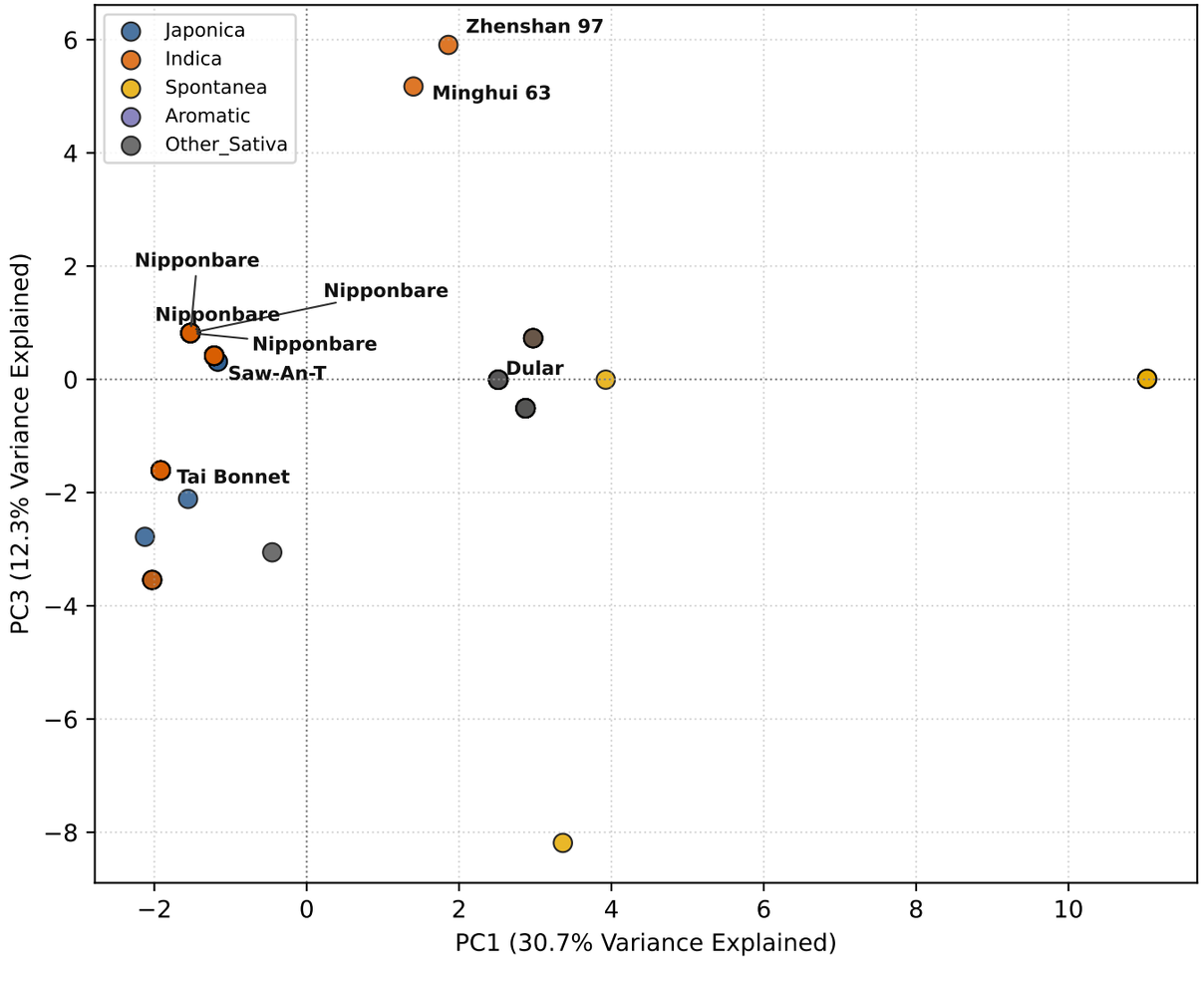

A. Scree Plot of Eigenvalues & Cumulative Variance Explained
Variance Explained (%)
Cumulative Variance (%)
Principal Component (PC)


B. Population Structure: PC1 vs PC2 by Varietal Group
Japonica (n=51)
Indica (n=26)
Spontanea (n=5)
Aromatic (n=1)
Other_Sativa (n=7)
PC2 (19.2% Variance Explained)
PC1 (30.7% Variance Explained)


C. Geographic Stratification: PC1 vs PC2 by Agro-Zone
Northern Margin Russia (n=56)
East Asia (n=24)
South Southeast Asia (n=6)
Middle East (n=2)
Other Unspecified (n=2)
PC2 (19.2% Variance Explained)
PC1 (30.7% Variance Explained)


D. Fine-Scale Differentiation: PC1 vs PC3 with Anchor Cultivars
Japonica
Indica
Spontanea
Aromatic
Other_Sativa
Zhenshan 97
Minghui 63
Nipponbare
Nipponbare
Nipponbare
Nipponbare
Saw-An-T
Dular
Tai Bonnet
PC3 (12.3% Variance Explained)
PC1 (30.7% Variance Explained)